CoRe: Construct-versus-Recall Routing for Knowledge Base Construction Under a Parameter Budget

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Abstract

We report a system for the LM-KBC shared task under a 32B budget on the total parameters of all inference-time components, which rules out scaling to a larger single model and caps any ensemble at the same 32B total. Starting from a rationalized chain-of-thought (CoT) pipeline that reached macro-F1 0.522 with Gemma-4-31B-it, we show that swapping to a smaller, differently-trained model (Qwen3.5-27B) under an identical pipeline recovers 24 percentage points of accuracy on a purely numeric relation (surface area), while costing accuracy on relations requiring broad factual recall. This shows that parametric knowledge coverage, not reasoning capability, is the limiting factor at this model scale. We then present four negative or reversed results (self-consistency sampling on numeric relations, a confabulation-guard prompt, reasoning-budget effects, and non-comparable validation splits) as diagnostic evidence for how scaffolding interacts with relation type. We introduce CoRe routing (Construct versus Recall), which assigns each relation to CoT or direct answering according to whether its answer must be constructed or merely recalled. CoRe reaches macro-F1 0.546 on validation and 0.541 on test.

1 Introduction

The LM-KBC challenge asks systems to construct knowledge base triples directly from a language model's parametric knowledge, given a subject entity and a relation, without external retrieval (Singhania et al., 2023; Petroni et al., 2019). Prior systems have shown that prompt phrasing, few-shot construction, and ensembling can move scores by 5–15 percentage points independent of the underlying model (Biester et al., 2023; Alivanistos et al., 2022). We study this sensitivity under an additional constraint that prior LM-KBC systems did not face: a hard limit of 32B on the total parameters of all inference-time components. This constraint matters because capacity cannot simply be added: scaling to a larger single model (Li et al., 2022) is ruled out, and any ensemble of models (Biester et al., 2023) must fit within the same 32B rather than stacking on top of a strong single model.

Our task instance covers six relations spanning three qualitatively different extraction problems: multi-valued list construction (*award recipients*, *country borders*), single-valued numeric recall (*venue capacity*, *geographic area*), and calibrated abstention (*place of death*, *stock exchange listing*, where a substantial fraction of gold answers are the empty set). We show that these three problem types respond to scaffolding in opposite directions, so no single global prompting strategy is optimal for all of them.

Our contributions are: (1) CoRe routing, a per-relation policy that outperforms any single global configuration we tested; (2) evidence that a model swap within the parameter budget recovers more accuracy on knowledge-limited relations than any prompting intervention we found; (3) four diagnosed negative results, each explaining why a plausible intervention does not transfer to this setting.

2 Task and Data

The task provides six relations over a shared entity pool, split into train/validation/test with approximately 100 rows per relation (68 for *countryLandBordersCountry* and 10 for *awardWonBy*). Two relations (*hasArea*, *hasCapacity*) are strictly single-valued numeric and never empty in gold; two (*personHasCityOfDeath*, *companyTradesAtStockExchange*) permit the empty set at substantial rates (39% and 36% of validation rows respectively); the remaining two permit variable-length lists. Scoring uses per-row precision/recall against a multi-alias gold set, macro-averaged across relations, with numeric answers matched under 5% relative tolerance.

3 System

3.1 Base pipeline

For each query we sample $k{=}5$ few-shot exemplars from the training split via a hash of (seed, subject, relation), which guarantees exemplar selection is independent of batch composition and processing order, a property we found necessary after observing that a shared random-number stream produced different exemplars for the same row across code revisions, confounding otherwise-controlled comparisons.

For relations assigned to the chain-of-thought (CoT) path (Wei et al., 2022) (§3.2), each exemplar's question is paired not with its gold answer directly but with a *rationalization*: the model is asked, separately and once per unique exemplar (cached across queries), to explain why the given answer follows from the question, treating the answer as already known. This produces a demonstration of reasoning style without requiring us to write relation-specific chains of thought by hand. The query itself is then answered with the same reasoning format, terminating in a `Final Answer:` line that we parse.

For relations assigned to the direct path, exemplars show the gold answer with no reasoning, and the query is answered directly under a low token budget.

Step 1: rationalize an exemplar

system: You are a careful research assistant. You will be given a factual question together with its correct answer. In 2–4 concise sentences, explain the reasoning that leads to this answer, as if working it out from scratch. Do not mention that the answer was provided to you.

user: Question: What is the total surface area of West Island in square kilometres... **Correct answer:** 6.23

→ *cached as this exemplar's rationale*

Step 2: answer the query

system: You are a knowledge base. First reason step by step, then on its own final line write 'Final Answer: ' followed by only the answer entities. Always commit to a specific value, even if you are unsure. Never answer None.

user: What is the total surface area of **West Island**...

assistant: West Island is the main island of the Cocos (Keeling) group. Sources give its area as about 6.2 square kilometres...
Final Answer: 6.23

...4 more exemplars...

user: What is the total surface area of **Bequia**...

Figure 1: The two-step prompt for a CoT relation, abbreviated. Step 1 runs once per unique exemplar and is cached; step 2 assembles the cached rationales into a few-shot context. The final sentence of the step-2 system prompt is the never-abstain policy, applied only to the two relations whose gold is never empty.

Relation	CoT	Direct
awardWonBy	**0.123**	0.078
hasArea	**0.610**	0.480
hasCapacity	**0.180**	0.160
countryLandBordersCountry	0.939	**0.978**
personHasCityOfDeath	0.380	**0.500**
companyTradesAtStockExchange	0.624	0.624

Table 1: Macro-F1 by relation under a global CoT vs. direct policy (Qwen3.5-27B, validation split). No single global choice is optimal.

3.2 CoRe routing

We evaluated CoT and direct answering as global choices before finding that neither dominates: routing every relation to CoT reached macro-F1 0.511 on validation; routing every relation direct reached 0.510; but choosing the better mode independently per relation reached 0.542. Table 1 shows the per-relation split. CoRe routing is built on this per-relation pattern. Relations whose answer must be *constructed*, such as award recipient lists or a two-step "is this company listed, and if so where" decision, benefit from explicit reasoning space. The numeric relations (*hasArea*, *hasCapacity*) also route to CoT, but for a narrower reason: a single reasoning pass helps the model lay out the arithmetic and commit to a well-formed value. What hurts these relations is not one reasoning pass but *additional* reasoning—self-consistency sampling and larger budgets (§4)—because their residual errors are memorization-bound (§6). Only *countryLandBordersCountry* and *personHasCityOfDeath*, where reasoning earns nothing over direct answering, take the direct path.

3.3 A confounded but instructive case: reasoning space for calibration

The *companyTradesAtStockExchange* tie in Table 1 initially suggested the relation was insensitive to reasoning. Inspecting raw model output revealed otherwise: under the direct path, the model's median output length was 4 characters. It emitted `"None"` immediately, with no room to execute the two-step decision our prompt asked for ("is this company publicly listed; if so, which exchange"). Rewriting the prompt to state the two-step decision explicitly, while leaving the relation on the direct path, raised macro-F1 from 0.624 to 0.653. Moving it to the CoT path so that the instruction had room to execute raised it further to 0.674, with the predicted-empty rate falling from 57 to 40 against a gold-empty rate of 36. The ordering matters: the prompt rewrite alone recovered less than half the gain, because a template's requirements and its assigned decoding path were mismatched. The template asked for reasoning the token budget could not support.

3.4 Never-abstain fallback

For the two strictly-numeric relations, an empty prediction always scores zero (gold is never empty), while any relation with non-empty gold rows scores identically whether the model abstains or guesses wrong. We therefore compute the median value of the training split at initialization and substitute it whenever an answer fails to parse, guaranteeing zero abstentions on these two relations.

4 Negative Results

We report four interventions that failed or reversed under direct measurement.

Self-consistency hurts single-valued numeric recall. Motivated by the intuition that sampling and aggregating should reduce variance, we applied self-consistency (Wang et al., 2023): we generated 5 completions per query at temperature 0.7 for *hasArea* on Gemma-4-31B-it (our earlier base model) and took the median. This broke 4 of the 37 rows that greedy decoding answered correctly while rescuing only 1, for a net loss (macro-F1 0.37 $\rightarrow$ 0.34). Manual inspection showed why: sampled completions for a given entity tend to *agree* with each other around a single misremembered value rather than scattering randomly around the truth. On Isle of Wight, greedy decoding correctly answered 384 km^2 (gold: 380.16); all 5 sampled completions clustered near 984. This indicates the model's errors on this relation are confident but incorrect recalled values rather than uncertain guesses, a failure mode that resampling cannot correct because it is not variance-driven.

Guarding against confabulation can regress accuracy. For *personHasCityOfDeath*, we observed the model conflating a person's city of death with their birthplace, their country's capital, or the city where they were famous. We rewrote the prompt to explicitly forbid these substitutions. Measured on the full validation run, macro-F1 fell from 0.500 to 0.470, driven by predicted-empty rate rising from 63 to 79 against a gold-empty rate of 39: the guard suppressed correct low-confidence answers along with incorrect ones. We revert to the original template.

Token budget effects are confined to the construct-type relations. We swept the CoT reasoning budget over 300, 500, and 2000 tokens on the test split, holding batch composition and every other setting fixed (Table 2). Two results stand out, and the second corrects an expectation we held during development.

First, the gains are confined to exactly the two construct-type relations on the CoT path. *awardWonBy* improves by 1.4pp and *companyTradesAtStockExchange* by 2.2pp from 300 to 2000 tokens, which is the direction Table 1 predicts: a longer budget helps when the answer has to be assembled rather than recalled. Almost all of the *companyTradesAtStockExchange* gain arrives between 500 and 2000 tokens, consistent with the median-output-length finding in §3.2 that this relation needs room to execute a two-step decision before it commits to an answer.

Second, the other four relations are numerically *unchanged* across all three budgets, identical to four decimal places in precision, recall, and F1 alike. This includes *hasArea* and *hasCapacity*, which CoRe assigns to the CoT path and which therefore did receive the larger allowance. We had expected the extra reasoning space to hurt them, by the same mechanism as the self-consistency failure above, and it does not. The explanation is a property of greedy decoding: the token at each step is the argmax over the prefix, so any query that already emitted its `Final Answer:` line within 300 tokens produces a byte-identical continuation at 500 or 2000. A larger budget can only change queries that were previously truncated mid-reasoning, of which these relations had few. Raising the reasoning budget under greedy decoding is therefore a one-sided operation: it can recover truncated answers but cannot disturb answers that already completed, which makes it safe to tune on the construct-type relations without re-measuring the rest.

The aggregate effect is correspondingly small: 0.5359 to 0.5408 macro-F1 across a nearly sevenfold increase in the token budget, at a proportional increase in decoding cost. We report the 2000-token configuration as our primary test result, but note that 300 tokens recovers 99.1% of its

Relation	Max new tokens 300	500	2000
CoT path			
awardWonBy	0.1524	0.1560	**0.1659**
companyTradesAtStockExchange	0.6833	0.6833	**0.7050**
hasArea	0.5600	0.5600	0.5600
hasCapacity	0.1327	0.1327	0.1327
Direct path			
countryLandBordersCountry	0.9657	0.9657	0.9657
personHasCityOfDeath	0.5100	0.5100	0.5100
All relations	0.5359	0.5360	**0.5408**
Zero-object cases*	0.7250	0.7250	**0.7273**

Table 2: Test-split macro-F1 by relation across CoT reasoning budgets under CoRe routing (Qwen3.5-27B). Only the two construct-type relations (*awardWonBy*, *companyTradesAtStockExchange*) respond to the budget. *hasArea* and *hasCapacity* are on the CoT path but already emit their answer within 300 tokens, so greedy decoding gives identical output at every budget; the two direct-path relations never receive the CoT allowance. *Zero-object cases are the subset of queries whose gold answer is the empty set, scored separately.

macro-F1, so the budget is not where the remaining headroom lies.

Small validation splits are not interchangeable with full-run scores. Twice during development we measured a relation in isolation (to reduce iteration cost) and compared the result against a baseline drawn from a full run over all six relations, concluding an intervention had helped or hurt when it had not. For *hasCapacity*, the full-run baseline (0.180) and the isolated-relation baseline (0.150) differ by 3 percentage points under an *identical* configuration; for *personHasCityOfDeath*, the gap was 5 points (0.500 full-run vs. 0.450 isolated). We traced this to batch composition: with greedy decoding, left-padding length depends on the other rows sharing a decoding batch, which depends on which relations are active in a given run. Moving one relation between reasoning paths shifted *other, untouched* relations' scores by up to 5 points purely through this channel. We recommend that KBC systems evaluated with batched greedy decoding treat cross-run relation-level comparisons as valid only when batch composition is held fixed, and report this as a source of measurement noise independent of the intervention under test.

5 Results

Table 3 compares our staged configurations against the model-swap baseline. The largest gain in our study came from changing the base model within the parameter budget rather than from any prompting intervention: under an otherwise-identical rationalized-CoT pipeline, Gemma-4-31B-it answered 37/100 *hasArea* queries within the 5% numeric tolerance, against 61/100 for Qwen3.5-27B. This 24-point swing exceeds the combined effect of every prompting intervention we measured on any single relation, and is consistent with our finding that errors on this relation are memorization failures rather than reasoning failures: a different model has different memorized facts, but no amount of reasoning scaffolding recovers a fact neither model stores.

Configuration	Macro-F1
Gemma-4-31B-it, all-CoT (baseline)	0.522
Qwen3.5-27B, all-CoT	0.511
Qwen3.5-27B, all-direct	0.510
Qwen3.5-27B, CoRe routing	0.542
+ rewritten *company* prompt	**0.548**

Table 3: Validation macro-F1 across pipeline configurations. CoRe routing outperforms either global policy under the same model, and the model swap alone is responsible for the majority of the gain over the original baseline on knowledge-bound relations. All figures here are validation. On the test split the same pipeline with *company* on the CoT path scores 0.5408 (Table 2); we did not measure the direct-*company* variant on test, so the two splits are not directly comparable row for row.

6 Error Analysis

Numeric relations degrade with entity obscurity

Accuracy on *hasArea* is almost monotonic in the size of the entity being asked about (Table 4). Every country-scale entity is answered within tolerance; small islands and lakes are answered correctly less than half the time. Since the arithmetic involved is identical across buckets, and the prompt states the unit conversions explicitly, this gradient is not a reasoning or formatting effect. It tracks how often the entity is likely to have appeared in pretraining.

Inspecting the reasoning traces confirms this. On the failures the model does not hedge or refuse; it commits to a specific figure and then reasons fluently about its precision. For Hopen it answers 1130 km^2 against a gold value of 47, noting that the figure "satisfies the requirement for three significant figures"; for Gogland it answers 142.5 against 21, again arguing about significant figures rather than about the island. For Donauinsel it recalls an area in hectares, converts correctly using the factor

Gold area (km^2)	n	Within 5%
<10	13	38%
10–100	21	43%
100–1k	29	66%
1k–100k	33	73%
>100k	4	100%

Table 4: *hasArea* accuracy by the magnitude of the gold value (Qwen3.5-27B, validation). Larger entities are more prominent in pretraining and are answered far more reliably, though the extraction problem is identical.

supplied in the prompt, and arrives at 10.0 against a gold value of 3.9: the arithmetic is right and the remembered quantity is wrong. These are retrieval errors dressed in correct procedure, which is why the scaffolding interventions in §4 could not touch them, and why the model swap in §5 could.

hasCapacity shows the same structure more severely: the venues in this split are largely high-school, minor-league and small-college grounds, and the model routinely produces a confident, precisely-stated figure for a venue it has evidently confused with another. For "Melaleuca Field in Idaho", a minor-league baseball park seating 3400, it reports that the total maximum capacity "is officially published as 16,112".

Abstention is already near-optimal

Three relations permit an empty gold set, and the scoring rule makes the trade-off asymmetric: on a row whose gold is non-empty, abstaining and guessing wrong both score zero, whereas on a row whose gold is empty, any guess scores zero where abstaining would have scored one. Guessing is therefore free on non-empty rows and expensive on empty ones.

This asymmetry makes it tempting to push the model toward answering more often, and we tried it. The break-even accuracy required is higher than it appears. Converting the w rows where the model wrongly abstains yields $w \cdot a$ points at answering-accuracy a, but any prompt change general enough to do so also disturbs the c rows where it abstains correctly. Break-even requires $a > c/w$. On *personHasCityOfDeath* the model answers with 41% accuracy against $c/w = 36/27$, so an intervention would need 133% accuracy to pay for itself; on *companyTradesAtStockExchange* it answers with 87% accuracy against $c/w = 32/25$, requiring 128%. On both relations the operation cannot break even at any accuracy the model could attain. Our confabulation-guard experiment (§4) is the empirical version of this calculation: it moved abstention in the intended direction and lost three points.

We take this as evidence that the abstention behaviour of an instruction-tuned model on this task is already close to a local optimum, and that the remaining headroom on these relations is in *which* rows it abstains on rather than how often. Improving that requires better-calibrated knowledge, not a better-worded instruction.

7 Related Work

Biester et al. (2023) apply prompt ensembling to Llama2 models at 7B/13B/70B scale, measuring a consistent 4.7–6.1 point gain from ensembling over the single best prompt at every model size, comparable in magnitude to a roughly 10x increase in parameter count in their setting. Their relation-level results include *PersonHasPlaceOfDeath*, directly comparable to our *personHasCityOfDeath*: they report a 14-point F1 swing (0.444 vs. 0.303) between two phrasings of the same question, and describe a fact-probing step for death-related relations that first asks whether the subject is deceased before eliciting a location, a two-step decomposition we independently arrived at for the same relation via direct error analysis, which we take as convergent evidence that this decomposition is a property of the task rather than of either system. Alivanistos et al. (2022) similarly decompose relation extraction into multi-step prompting chains, and report that the empty-answer case is a distinct failure mode requiring its own treatment rather than falling out of the extraction step, which matches what we find in §6.

Prompting versus training. Li et al. (2022) took a different approach entirely, using task-specific pretraining rather than prompting, and won Track 1 of the original ISWC 2022 challenge. Zhang et al. (2023) likewise treat the problem as knowledge engineering over Wikidata rather than as prompt design. Both are reminders that our findings are specific to the prompting-only regime imposed by this task's parameter constraint, not a claim that prompting is the best available lever in general. Notably, Biester et al. (2023) report that fine-tuning *underperformed* their non-fine-tuned model in the same prompt-ensembling setup (0.562 vs. 0.589), so the ordering between the two approaches appears to depend on the model and the budget rather than being fixed.

Knowledge as the limiting factor. The framing of language models as knowledge bases originates with Petroni et al. (2019), whose LAMA probe assumed single-token, single-answer relations. The shared task relaxes both assumptions, and our results suggest that the harder setting exposes where the analogy breaks down: a knowledge base can be queried for an obscure entity as reliably as for a famous one, whereas the model's accuracy on *hasArea* falls from 100% to 38% as entity prominence decreases (Table 4) with no change in the extraction problem itself. This is consistent with reports that recall of long-tail facts scales with pretraining frequency and parameter count more than with inference-time technique (Kandpal et al., 2023; Mallen et al., 2023), and it is the reason we treat the 32B ceiling, rather than our prompting choices, as the binding constraint on this system.

8 Conclusion

We built a knowledge base construction system under a 32B budget on total inference-time parameters, which rules out a larger single model and caps any ensemble at the same 32B total. Within that constraint we found that no single prompting policy is optimal across relations: the better of chain-of-thought and direct answering is relation-specific, and routing on it (CoRe) outperforms either global policy by three points.

The more useful finding for future systems may be the negative one. Across a dozen interventions, everything that operated on the scaffolding around the model, including self-consistency sampling, exemplar selection, reasoning budget, and several prompt rewrites, moved the score by less than the run-to-run variation induced by batch composition. The one change that moved it substantially was replacing the base model with a different one of comparable size. On *hasArea* that swap was worth 24 points where no prompting change was worth three. We read this as evidence that at this scale the task is limited by which facts a model has memorised rather than by how effectively those facts are elicited, and that reporting practice should account for measurement noise at the few-point level before attributing gains to technique.

9 Limitations

Our routing decisions were tuned on a validation split of under 500 rows total (10–100 per relation), and §4 documents that even same-configuration reruns can differ by several points due to batch-composition effects; some of the per-relation deltas we report as evidence for routing decisions are close to this noise floor and should be treated as directional rather than precise. Our self-consistency finding is demonstrated on two numeric relations from one task instance; we do not verify it holds for numeric relations in general. The reasoning-budget sweep in Table 2 is a single run per budget with no seed variance, and its two non-zero deltas (1.4pp and 2.2pp) are within the range that §4 shows batch composition alone can produce, so the ranking of the three budgets should be read as directional rather than established. The 32B constraint that motivates this study is specific to our shared-task track and may not transfer to settings without it. Finally, our model-swap finding (different models recover different facts) suggests that an ensemble of two or more models sharing the 32B budget might outperform routing within a single model. The task rules permit such within-budget ensembles; we did not test one, and see it as the most direct follow-up to this work.

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A Full Score Tables

Table 5 gives per-relation macro-F1 for every configuration in the CoRe lineage on the validation split, and Table 6 places those numbers beside two earlier experimental lineages that we ran on the same task with different pipelines.

The cross-lineage comparison carries an important caveat. The three groups in Table 6 differ in more than their base model: the Gemma sampling lineage draws 20 (or 5) samples per numeric query and routes four string relations through a yes/no self-consistency gate, and the Qwen3-32B and Qwen3.5-9B lineages use a different codebase again, with relation-specific numeric prompting tiers. Differences across groups therefore confound model identity with pipeline design and should not be read as evidence about either one alone. Comparisons within a group are controlled and interpretable; comparisons across groups are reported only for completeness.

Two entries are worth isolating because they bear on the paper's central claim. Gemma-4-31B and Qwen3.5-27B under the *identical* CoRe pipeline (rows 1 and 2 of Table 5) differ by 24 points on *hasArea* (0.370 vs. 0.610) while moving in the opposite direction on *companyTradesAtStockExchange* (0.774 vs. 0.624). This is the controlled version of the model-swap result reported in §5. Separately, the Gemma sampling lineage reached 0.755 on *companyTradesAtStockExchange* with none of our prompt work, close to the 0.774 Gemma reaches under CoRe and well above every Qwen configuration, which suggests abstention behaviour on that relation is driven more by the base model than by the scaffolding around it.

	award	company	borders	area	capacity	death
Gemma CoT	.150	.774	.980	.370	.200	.470
Qwen all-CoT	.123	.624	.939	.610	.180	.380
Qwen all-direct	.078	.624	.978	.480	.160	.500
Qwen CoRe	.119	.624	.978	.610	.180	.500
+prompt	.119	.653	.978	.610	.180	.500
+prompt+CoT	.123	.674	.978	.610	.180	.470

Table 5: Per-relation macro-F1, CoRe lineage, validation split. Overall macro-F1: Gemma CoT 0.522, Qwen all-CoT 0.511, Qwen all-direct 0.510, Qwen CoRe 0.542, *+prompt* (rewritten *company* template, relation left on the direct path) 0.548, *+prompt+CoT* (same template, relation moved to the CoT path) 0.546. The last two differ only in which path *company* takes: moving it to CoT gains 2.1pp on that relation but loses 3.0pp on *personHasCityOfDeath*, which shares its batch group. Rows 1 and 2 share an identical pipeline and differ only in base model.

System	Lineage	Overall
Qwen3.5-27B +prompt	CoRe	**0.548**
Qwen3.5-27B +prompt+CoT	CoRe	0.546
Qwen3.5-27B CoRe	CoRe	0.542
Gemma-4-31B CoT	CoRe	0.522
Qwen3.5-27B all-CoT	CoRe	0.511
Qwen3.5-27B direct	CoRe	0.510
Gemma 4 31B n=5	Gemma sampling	0.507
Gemma 4 31B n=20	Gemma sampling	0.466
Tier-1 units	Qwen3-32B	0.454
Official baseline	Qwen3-32B	0.446
Agentic multi-step	Qwen3-32B	0.403
CoT (hasArea-only)	Qwen3.5-9B	0.420
Baseline	Qwen3.5-9B	0.407

Table 6: Overall macro-F1 across three experimental lineages on the validation split, best two or three configurations per lineage. Lineages differ in pipeline as well as model (see text), so cross-lineage rows are not a controlled comparison.

B Reproducibility

Our code, configurations, and prompt templates are available at `https://github.com/MightBeFonias/CoRe`. All results use greedy decoding with k=5 few-shot exemplars sampled by a hash of (seed, subject, relation) with seed 42, so exemplar selection is reproducible and independent of batch composition. The CoRe configuration assigns *hasArea*, *hasCapacity*, *awardWonBy* and *companyTradesAtStockExchange* to the CoT path and the remaining two relations to the direct path, with a 2000-token reasoning budget on the CoT path and 64 tokens on the direct path. Numeric relations fall back to the training-split median when an answer fails to parse. Batch size is 4 throughout; §4 documents why this value must be held fixed across runs that are to be compared.